

INHA UNIVERSITY IN TASHKENT
School of Computers and Information Engineering
Fall semester 2024

Computer Algorithms (CIE3090)
Assignment #2: (Solving recurrence Equations)

Submission Date: 29 /09/ 2024 (23:30 Hr) Sunday

NOTE : Solve the following question with pen(blue or black)

1. Computer typed solution are not acceptable (only handwritten allowed)
2. Check the format of assignment- submission at last page.
3. Submissions not conforming to the given format will not be evaluated.

Q1. Solve recursive Equations using back Substitution method

a. $T(n) = c + 2 T(n/2) \quad : n > 1$ $\quad \quad \quad = 1 \quad \quad \quad : n = 1$	b. $T(n) = n + 2 T(n-1) \quad : n > 1$ $\quad \quad \quad = 1 \quad \quad \quad : n = 1$
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Q2. Solve recursive Equations using recurrence Tree

a. $T(n) = 2 T(n/2) + n \quad : n > 1$ $\quad \quad \quad = 1 \quad \quad \quad : n = 1$	b. $T(n) = 2 T(n-1) + 1 \quad : n > 1$ $\quad \quad \quad = 1 \quad \quad \quad : n = 1$
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Q3 Solving Recurrence Equation using Masters Theorem

$$T(n) = a T(n/b) + \Theta(n^k \log^p n)$$

$a \geq 1, b > 1, k \geq 0$ are constants and p is real number

For each question, identify the value of a , b , p , and k . Mention the case applicable and write the answer.

Sample Solution

$$T(n) = 8 T(n/2) + n^2$$

$a=8, b=2, k=2, p=0$ [since $a > b^k$], therefore case 1 applicable

Therefore $\Rightarrow T(n) = \Theta(n^{\log_b a}) = \Theta(n^3)$

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THIS SHEET TO BE SUBMITTED
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ID _____

Name _____

Solution 1a	Solution 1b

Solution 2a	Solution 2b

3a. $T(n) = 4T(n/2) + n^2$ [a= __, b= __, k= __, p= __] Case = _____ Ans = _____	3b. $T(n) = 2T(n/2) + n \log n$ [a= __, b= __, k= __, p= __] Case = _____ Ans = _____	3c. $T(n) = 2T(n/2) - n \log n$ No Solution, give Reason
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